

Model	k	Data Format	Overall	Instruction Retrieval	Begin	Middle	End
Claude3.7-Sonnet	2	f_1	0.4310	0.8943	0.4481	0.4074	0.4359
Claude3.7-Sonnet	2	f_2	0.4215	0.9071	0.4122	0.4366	0.4166
Claude3.7-Sonnet	2	f_3	0.4205	0.8880	0.4191	0.4179	0.4253
Claude3.5-Sonnet-V2	2	f_3	0.4182	0.8765	0.4214	0.4126	0.4204
Claude3.7-Sonnet	1	f_2	0.3948	0.8267	0.3878	0.3834	0.4161
Claude3.5-Sonnet-V2	2	f_1	0.3941	0.8721	0.4036	0.3904	0.3862
Claude3.5-Sonnet-V2	2	f_2	0.3939	0.8631	0.4005	0.3956	0.3838
Claude3.7-Sonnet	1	f_1	0.3932	0.8205	0.3959	0.3895	0.3939
Claude3.7-Sonnet	1	f_3	0.3869	0.8163	0.3932	0.3821	0.3843
Claude3.5-Sonnet-V1	2	f_3	0.3716	0.8304	0.3750	0.3730	0.3660
Claude3.5-Sonnet-V1	2	f_1	0.3485	0.7830	0.3533	0.3490	0.3419
Claude3.5-Sonnet-V1	2	f_2	0.3383	0.7910	0.3437	0.3259	0.3452
Claude3.5-Sonnet-V2	1	f_1	0.3312	0.7311	0.3270	0.3333	0.3341
Qwen3-14B	2	f_1	0.3279	0.8419	0.1668	0.2880	0.5728
Claude3.5-Sonnet-V1	1	f_3	0.3187	0.7057	0.3185	0.3342	0.3018
Qwen3-14B	1	f_1	0.3122	0.7792	0.1645	0.2828	0.5289
Claude3.5-Sonnet-V1	1	f_1	0.2963	0.6491	0.3019	0.2911	0.2951
Qwen3-14B	2	f_2	0.2961	0.8418	0.1250	0.2623	0.5468
Qwen3-14B	1	f_2	0.2845	0.7713	0.1142	0.2584	0.5256
Qwen3-14B	2	f_3	0.2712	0.7879	0.1130	0.2248	0.5198
Gemma3-27B-IT	2	f_2	0.2559	0.7559	0.0755	0.2392	0.4990
Claude3.5-Sonnet-V2	1	f_3	0.2543	0.5152	0.2604	0.2606	0.2396
Gemma3-27B-IT	2	f_1	0.2528	0.7283	0.0809	0.2414	0.4797
Llama3.3-70B	2	f_1	0.2497	0.6612	0.2620	0.2366	0.2488
Gemma3-27B-IT	2	f_3	0.2362	0.6717	0.07314	0.2209	0.4561
Llama3.3-70B	2	f_2	0.2362	0.5924	0.2318	0.2327	0.2454
Gemma3-27B-IT	1	f_1	0.2221	0.6138	0.0770	0.2144	0.4113
Qwen3-14B	1	f_3	0.2110	0.6368	0.0681	0.1887	0.4137
Claude3.5-Sonnet-V2	1	f_2	0.2084	0.4654	0.2105	0.2044	0.2102
Llama3.3-70B	2	f_3	0.2065	0.5376	0.2020	0.2013	0.2179
Claude3.5-Sonnet-V1	1	f_2	0.2048	0.5038	0.2051	0.2026	0.2068
Gemma3-27B-IT	1	f_2	0.1847	0.5225	0.0557	0.1699	0.3616
Gemma3-27B-IT	1	f_3	0.1740	0.4960	0.0526	0.1695	0.3303
Llama3.3-70B	1	f_2	0.1447	0.2503	0.1463	0.1477	0.1393
Llama3.3-70B	1	f_1	0.1358	0.2042	0.1374	0.1338	0.1360
Llama3.3-70B	1	f_3	0.1299	0.2444	0.1413	0.1268	0.1191

Table 31: Task 1 accuracy of top $k = 1$ and $k = 2$ instructions retrieval and next agent action prediction (only English conversations). Results are shown across different models and the three complex instruction formats (f_1 , f_2 , f_3).

Model	Approach 1			Approach 2		
	f_1	f_2	f_3	f_1	f_2	f_3
Claude3.7-Sonnet	0.7608	0.7525	0.7528	0.7298	0.7346	0.7298
Claude3.5-Sonnet-V2	0.7215	0.7147	0.7238	0.6874	0.6826	0.6847
Claude3.5-Sonnet-V1	0.6847	0.6930	0.6945	0.6854	0.6771	0.6748
Llama3.3-70B	0.7283	0.7283	0.7159	0.5636	0.5747	0.5368
Qwen3-14B	0.7046	0.6975	0.6948	0.6937	0.6910	0.6811
Gemma3-27B-IT	0.6968	0.6857	0.6688	0.5616	0.5454	0.5396

Table 32: Task 2 accuracy for English conversations across approaches and complex instruction formats. The best result per column is **bolded** and the best result per model-approach pair is underlined.

Model	f_1	f_2	f_3
Claude3.7-Sonnet	0.9529	0.9574	0.9565
Claude3.5-Sonnet-V2	0.9532	0.9498	0.9471
Claude3.5-Sonnet-V1	0.9254	0.9356	0.9251
Gemma3-27B-IT	0.8952	0.8734	0.8695
Qwen3-14B	0.2269	0.2208	0.2178
Llama3.3-70B	0.1435	0.1610	0.1511

Table 33: Task 3 compliance rate across models and complex instruction formats (only English conversations). The best result per complex instruction format is **bolded** and the best result per model is underlined.